Contraception & Abortion Pharmacology

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Block: Endocrine/Urology/Reproduction

Thread: Pharmacology

Conflicts of Interest: None

Learning Objectives

By exam time, you should be able to:

1. Compare and contrast the mechanisms of action, signature adverse effects, and contraindications of combined hormonal (estrogen-progestin) contraceptives.
2. Compare and contrast the different modes of contraceptive delivery (oral, injectable, implantable, intrauterine, vaginal ring, transdermal patch).
3. Describe the rationale for the use of progestin-only contraceptives and long-acting progestins.
4. Classify the agents used for emergency (postcoital) contraception by their MOA and appropriate time frame for use.
5. Describe the mechanisms of action and signature adverse effects of abortion medications.

Study Questions

1. Mary is a 45-year-old woman with two children who is interested in contraception. She smokes cigarettes and has a past medical history of high blood pressure. She desires to bleed every month. What contraceptive options would be good for Mary?

2. Rae is a 30-year-old woman with one child. She has no medical problems. She missed her period and is seen in the office and is found to be 8 weeks pregnant. Her hemoglobin is 10.5. Is Rae a candidate for medication abortion? Why or why not?

3. Jasmine had unprotected intercourse two days ago and wishes to prevent a conception. What is the most effective form of emergency contraceptive for Jasmine to use? How does each form of emergency contraception work?

Introduction

Contraception at a glance

1) Ideal Contraceptives

a) 100 % effective

b) 100 % non-toxic

c) No coitus-related activity required

d) Reversible

e) Inexpensive

f) Simple to use

g) Acceptable to all groups

2) Contraceptive Efficacy

a) **Pearl Index** – The number of failures per 100 woman-years of exposure. Limitation: this index does not describe the effect of duration of use on efficacy. It is a cross-sectional population statistic

b) **Life-Table Analysis** – This method is based on the failure rate for each month of method use. Data from patients who drop out of a study before completion can thus be included. This statistical technique is generally applied to measurement of survival in oncologic studies.

c) **Method-Effectiveness** – This is the efficacy with correct (ideal) use of the contraceptive method. It is usually determined in studies on highly motivated and closely supervised patients. This is also known as perfect use.

d) **Use-Effectiveness** – This is the efficacy when the method is in normal unsupervised use by unselected patients. This represents the "real world." This is also known as typical use.

CONTRACEPTIVE FAILURE RATES

% of women who will become pregnant during the first year of use, by method

Method	Typical use	Perfect use
Implant	0.1	0.1
Vasectomy (male sterilization)	0.15	0.1
IUD (hormone-releasing)	0.1–0.4	0.1–0.3
Tubal surgery (female sterilization)	0.5	0.5
IUD (copper T)	0.8	0.6
Injectable	4	0.2
Combined and progestin-only pill	7	0.3
Vaginal ring	7	0.3
Patch	7	0.3
Male condom	13	2
Sponge (when used by women who have not given birth)	14	9
Fertility awareness–based methods*	2–34	<1–12
Diaphragm (with spermicidal cream or jelly)	17	16
Withdrawal	20	4
Internal (female) condom	21	5
Spermicides	21	16
Sponge (when used by women who have given birth)	27	20
No method	85	85

*Range of estimates comes from a small number of moderate-quality studies and may not apply to all populations; higher-quality data are needed (Peragallo Urrutia et al., 2018). *Notes:* Typical-use failure rates express effectiveness among all women who use the method, including those who use it inconsistently and incorrectly. Perfect-use failure rates express effectiveness among only those women who use the method both consistently and correctly. IUD=intrauterine device.

Source: Guttmacher.org

3) Components of hormonal contraceptives

Combined estrogen and progestin; Progestin only

Synthetic estrogens: Ethinyl estradiol; mestranol;

estradiol valerate; estradiol cypionate; estetrol

Progestins (synthetic progesterone):

Testosterone derivatives (19C): Estranes, gonanes

Progesterone derivatives (21C): pregnanes

Spirolactones: Drosperinone

4) How does contraception work?

Combined estrogen and progestin: Ovulation inhibitors

Progestin only:	Cervical mucus thickeners
	Ovulation inhibitors
Non-hormonal:	Spermicidal
	Barrier

5) How do the ovulation inhibitors work?

Inhibition of gonadotropin secretion

Progestins: suppress LH release; atrophy of endometrial glands

Estrogen: inhibits FSH release; increases progesterone receptor concentration; stabilizes endometrium

6) Combined hormonal contraception

Different formulations of combined hormonal methods

Overall mechanism of action:

i) Suppress ovulation by preventing LH surge

ii) Thicken cervical mucus

iii) Decrease tubal motility

Combined oral contraceptives

a) Can have varying doses of estrogen component
b) Can have varying doses of progestin components
c) Take daily around the same time every day
d) More than 80 different brands

e) Usually 3 weeks active medication and one week of placebo

Delivery Modalities:

- Oral contraceptives (combined oral contraceptives: COCs)
- Transdermal patch
- Vaginal rings

***All* combined oral contraceptives contain:**

- **A progestin**
- **An estrogen component**
    - Most with Ethinyl estradiol
    - One with estradiol valerate
    - One with Estetrol
- Daily use
- Preferably the same time every day
- 3 weeks with medication, one week without, withdrawal bleed (not "period") during the week without medication
- First pass metabolism: absorbed through GI tract, metabolized hepatically
- Failure rate
    - Typical use 7%
    - Perfect use <1%

Transdermal patches

- Norelgestromin patch

  One patch per week x 3 week, then one week off
  Avoids first pass metabolism
  Continuous delivery to systemic circulation
    - "Area under the Curve" is 60% higher than in a 35 mcg pill
    - Possible increase in estrogen side effects, such as VTE
- Less effective if body weight is >196 pounds for norelgestromin/ EE patch
- Levonorgestrel patch
  One patch per week x 3 week, then one week off
  Not approved for BMI> 30: lower efficacy, higher risk of DVT

Monthly vaginal ring

- Ethinyl estradiol (EE) and Etonorestrel (ENG)
  Worn for 3 weeks with a 1-week interval

Low estrogen dose
Continuous dosing
No first pass metabolism/ no nausea
Lowest amount of break-through bleeding

1-year vaginal ring

- Ethinyl estradiol (EE) and segesterone acetate (SE)

  One ring for a whole year of use

  Wear for 3 weeks, remove for one week

  Store ring for the ring-free week; replace the same ring

Pharmacokinetics of combined hormonal methods: can reduce overall exposure by using different delivery methods

7) Estrogen containing contraception

Risks of combined hormonal methods:

- Rare, but serious
- Cardiovascular/thrombotic events
  - Stroke
  - Hypertensive urgency/ emergency
  - Myocardial infarction
  - Thrombotic events: PE, DVT

    Risk decreases with duration of use and decreasing E dose

Consult CDC medical eligibility criteria for contraception to see who should/shouldn't use certain types

Contraindications to estrogen containing methods:

- History of thrombotic event/thrombosis
- Clotting disorder
- Hypertension
- Migraines with aura
- Tobacco use over the age of 35

Estrogen containing CHC increases VTE risk

Potential side effects of exogenous hormones: (slide 35)

Estrogen excess: Nausea, breast tenderness, headaches, fluid retention; dysmenorrhea, menorrhagia, uterine fibroid growth
Estrogen deficiency: Early cycle (days 1-9) breakthrough bleeding and spotting; absence of withdrawal bleeding.

Progesterone excess: Increased appetite, weight gain, acne, oily skin, hirsutism, depression, fatigue, irritability.

Progesterone deficiency: Heavy menses, dysmenorrhea, Late cycle (days 10-21), breakthrough bleeding/ spotting

Effect of combination OCs on Androgen pathways:
Increases liver production of sex hormone binding globulin, thus reduces circulating free testosterone. Also less testosterone synthesis. Fewer androgenic effects (acne, hirsutism, etc).

Clinical relevance:

- All CHC improve androgenic profile
- No demonstrated association between CHC and weight changes
- No demonstrated association between CHC and depression

8) Progestin-only contraception

How does it work?

- Ovulation inhibition
- Cervical mucus thickening

(varies by delivery route)

Contraceptive Implant-Nexplanon®

- Progestin only implant, approved 2006
- Single rod implant inserted under skin of upper inner arm
- Impregnated with 68mg of etonogestrel (a type of progestin)
- Initially releases 60-70 mcg of etonogestrel, then 35-45 mcg in the first year, 30- 40mcg at 2 years, then 25-30mcg at 3 years
- Duration: FDA approved for 3 years, but effective for 5 years
- Most effective method (>99%)
- Mechanism of Action: Nexplanon®

  i) Inhibition of

  ovulation

ii) Cervical mucus thickening iii) Suppresses endometrium

a) Advantages of implant

i) Reduction in dysmenorrhea

ii) 3-5 years of pregnancy prevention

i) Disadvantages of implant

i) Unpredictable irregular vaginal bleeding

Injectable contraception: Medroxyprogesterone acetate

Depot Medroxyprogesterone Acetate (the shot)

Depo-provera, DMPA

a) Mechanism of action

i) Inhibits ovulation

ii) Thickens cervical mucus

b) No estrogen

c) Injected in deltoid or gluteus once every 3 months

d) Low maintenance; long acting

e) Biggest side effect is irregular bleeding

f) Can cause weight gain

DMPA side effects

- 1st several months: unpredictable or prolonged spotting
- 40-50% have amenorrhea after 1 year
- 20-25% discontinue because of menstrual issues
- Transient/reversible bone mineral density loss (because of hypoestrogenic effects)
- ****This is the only contraceptive associated with weight gain.****

9) Cervical mucus thickeners: progestin only pills; progestin IUD

How does this work? The cervix is responsive to estrogen and progesterone. Estrogen stimulates cervical mucus to be thin and penetrable by sperm. Estrogen levels are high at time of ovulation and cervical mucus is thin. Progesterone thickens cervical mucus, which is protective for pregnancy. Steady progestin level ensures that cervical mucus stays thick, thus preventing sperm and possibly bacteria from entering the upper genital tract.

Progestin-only "mini pill" (Norethindrone; Norgetstrel)

- Mechanism: thickens cervical mucus
- Can be taken by patients with contraindications to estrogen
- Take continuously; must take at exact same time each day or decreased efficacy (3 hour window)
- Variable bleeding profile
- No pill-free interval
- Failure rates: 3%
- OPill (norgestrel) over the counter pill approved 7/2023

Drospirenone (4mg): more forgiving dosing window.

- Mechanism: Atrophy of endometrium; thickens cervical mucus
- Safe for people who cannot take estrogen
- Spironolactone derived so can cause hyperkalemia.

10) Hormonal Intrauterine Device (IUD)

a) Levonorgestrel intrauterine devices

T-shaped device containing varying levels of levonorgestrel

i) Most effective (>99%)

ii) Mechanisms of Action: local progestin effect

(1) Cervical mucus thickening

(2) Sperm motility and function inhibition

(3) Suppresses endometrium

(4) Does NOT inhibit ovulation

iii) Advantages of Levonorgestrel-IUD

(1) Protective against uterine hyperplasia

(a) May be associated with a reduced risk of endometrial cancer

(2) Delivers even less progestin that the progestin-only pill, with minimal side effects

(3) Improvement in heavy and/or painful menstrual bleeding

(4) May be associated with reduced risk of PID (pelvic inflammatory disease)

(5) Ease of use

iv) Disadvantages of Levonorgestrel IUD

May cause irregular bleeding and cramping for the first 3-6 months of use

Comparing the 5 IUDs

Name	Dose	Duration	Daily release	Amenorrhea rates
Liletta®	52 mg	8years	18.6 mcg/ 24hr	19% 1yr/40% 3 yr
Mirena®	52mg	8years	20mcg/24hr	20% 1yr/40% 3 yr
Kyleena®	19.5mg	5 years	17.5mcg/ 24hr	12% 1yr
Skyla®	13.5mg	3 years	14.5mcg/ 24hr	6% 1 yr

11) Non-hormonal contraception

Copper IUD; Vaginal gel

Intrauterine Device: Copper IUD

No hormonal component

a) Duration: FDA approved for 10 years, effective for 12 years

b) Most effective (>99%)

c) Mechanism of action:

i) Sterile inflammatory response

ii) Sperm function and motility inhibited iii) Decreased tubal motility

Can be used as EC after unprotected intercourse

Advantages

- No hormonal component
- FDA approved 10 years of contraception, clinical data support 12 years

Disadvantages:

- Heavier menstrual period
- More cramping during menstrual period

Contraindications to IUD use

- Uterine anomalies, e.g. fibroids, congenital anomalies
- Blood dyscrasias
- Undiagnosed abnormal uterine bleeding
- Acute PID or history of PID with risk factor
- Untreated acute vaginitis or cervicitis
- Pregnancy or suspicion of pregnancy
- Carcinoma of cervix or endometrium
- Wilson's disease or allergy to copper (for copper IUD)
- Genital actinomycosis

Vaginal Gel (Phexxi)

- 1.8% lactic acid, 1% citric acid, 0.4% potassium bitartrate
- Keeps vaginal pH between 3.5 and 4.5
- Inhibits sperm motility
- Use up to one hour before each act of sex
- Comes in prefilled individually wrapped applicator

12) Emergency Contraception

a) Taken as soon as possible after unprotected intercourse

b) Mechanism of Action

i) Inhibits LH surge and delays ovulation (oral EC)

ii) Inhibits sperm/tubal motility (IUD)

Does *not* appear to inhibit implantation

Does *not* interrupt or affect an established pregnancy v) Does *not* have any serious side effects or contraindications, excluding allergy

c) Different methods

i) Levonorgestrel (Plan B)

(1) Use within 3 days of unprotected intercourse

(2) Sooner is better/ more effective if taken ASAP

(3) Prevents LH surge

(4) Available without a prescription

(5) Decreases risk of pregnancy 75% during luteal phase

(6) Less effective if BMI >30

ii) Ulipristal acetate (ella)

(1) Selective progestin receptor modulator

(2) Prescription only

(3) Use within 5 days of unprotected intercourse

(4) One 30mg tablet

(5) Less effective for people with BMI >35

iii) Yuzpe method

High dose combined hormonal method

Rarely used

Less effective

More side effects

iv) Copper IUD

(1) Different Mechanism of action

(a) Kills sperm

(b) Inhibits tubal motility

(c) Place within 5 days of unprotected intercourse

(d) Most effective form of emergency contraception

(e) 52 mg LNG IUD (Mirena, Liletta) are non-inferior to Copper IUD

13) Medication Abortion (MAB)/ Medical Management of Miscarriage

Medication vs. procedural

a) Mifepristone

i) RU-486

ii) Derived from norethindrone

iii) Selective progestin receptor modulator (SPRM)

iv) Binds more strongly to the progesterone receptor and inactivates it

v) Causes decidual necrosis and cervical softening/ripening

vi) Dose is 200mg

Transport protein is actually saturated at 100mg

vii) Reaches peak serum concentration in 2 hours

viii) Half-life is 24-29 hours

b) Misoprostol

i) Prostaglandin analog (PGE1 agonist)

ii) Causes uterine contractions and cervical dilation

iii) Inexpensive

iv) Stable at room temperature

iv) Side effects: Nausea/Vomiting/Diarrhea; fever, uterine cramping

c) Medication abortion regimens

i) Mifepristone 200mg

ii) Misoprostol 800mcg

(1) Buccally, 24-48 hours after mifepristone

iii) Up to 10-weeks gestation

iv) Combination causes bleeding and expulsion of the pregnancy

v) Confirm pregnancy expulsion

Contraindications: anemia; bleeding disorder/ anticoagulated; chronic steroid use/ adrenal failure; prophyria

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. **Mary is a 45-year-old woman with two children who is interested in contraception. She is a smoker and has a past medical history of high blood pressure. She desires to bleed every month. What contraceptive options would be good for Mary?**

Answer: Since Mary is 45 years old, uses tobacco, and has a hypertension she is not a candidate for combined hormonal contraception (please consult the CDC medical eligibility criteria for contraception). If she would like to menstruate every month the most effective method of birth control for her is the copper IUD. If she did not want to bleed every month or would accept irregular bleeding she could use the Nexplanon, hormonal IUD, depo-provera injection, and progesterone only pill. Non-medication options for Mary would include condoms, diaphragm, or permanent sterilization.

2. **Rae is a 30-year-old woman with one child. She has no medical problems. She missed her period and is seen in the office and is found to be 8 weeks pregnant. Her hemoglobin is 10.5. Is Rae a candidate for medication abortion? Why or why not?**

Answer: Rae is an excellent candidate for medication abortion. She is within 10 weeks of pregnancy, is not significantly anemic, and has no medical issues. She is also an excellent candidate for procedural abortion.

3. **Jasmine had unprotected intercourse two days ago and wishes to prevent a conception. What is the most effective form of emergency contraceptive for Jasmine to use? How does each form of emergency contraception work?**

Answer: The IUD is the most effective emergency contraceptive option and can be used for 5 days after unprotected intercourse. Historically, Copper IUD was the recommended type for emergency contraception, but recent studies have shown that the levonorgestrel IUD (Mirena or Liletta) is non-inferior to Cu IUD for emergency contraception. The table below summarizes emergency contraceptive options. Remember that emergency contraception is not an abortifacient – it functions to prevent fertilization.

Table 1

Major Methods of Emergency Contraception (EC), Availability and Policy in the U.S.

Brand Name	Efficacy	Timing after intercourse	Availability in the U.S.
Progestin-only pills: *Plan B One Step* and other brands	81 - 90% reduced pregnancy risk	Within 72 hours	One-dose versions approved for availability "over the counter" without age restrictions.
Ulipristal acetate: *ella*	2.1% failure rate; reduced pregnancy risk is 65% lower than when using progestin-only pills	Within 120 hours	Prescription only
Combined/Combination pills (estrogen and progestin): many brands	75% reduced pregnancy risk	Within 120 hours	Prescription only
Copper IUD: *Paragard*	99% reduced pregnancy risk	Within 120 hours	Requires clinician visit
Levonorgestrel IUD (52 mg): *Mirena* and *Liletta*	0.3% failure rate	Within 120 hours	Requires clinician visit

Note: All percentages presented are approximations. Conflicting research on efficacy of progestin-only pills and ulipristal acetate for women who are overweight and/or obese.

Source: The American College of Obstetricians and Gynecologists. Emergency Contraception. Last Updated November 2021; Planned Parenthood, IUD. Accessed August 2018; Trussell J, Raymond E, & Cleland K. Emergency Contraception: A Last Chance to Prevent Unintended Pregnancy. February 2018. Turok D.K. et al. Levonorgestrel vs. copper intrauterine devices for emergency contraception. New England Journal of Medicine. January 2021.

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